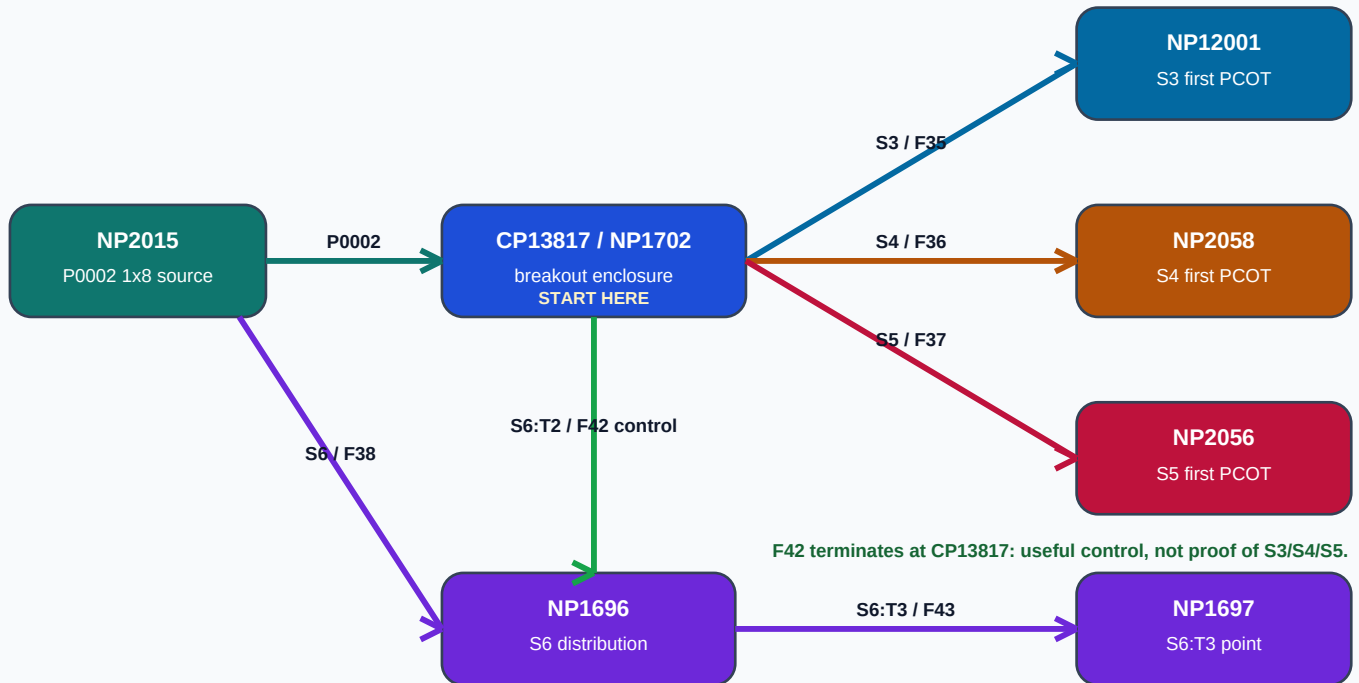


Node 54 / P0002 Field Troubleshooting Print

Ruidoso Revisit - field sequence built from the project staking KMZ and the Node 54 diagnostic model. Use this as a test plan, not as an as-built or authority to change plant.

IMPORTANT: The amber notes below are field hypotheses. A poor reading does not prove a splice is bad until the same identified fiber is checked immediately before and after the suspected interval with clean test setup.

Engineering relationship from the staking KMZ. Diagram is schematic only; it is not a surveyed route.



At each PCOT: identify the physical incoming and outgoing ports before comparing readings.

Start point and control

Start at CP13817 / NP1702. The model puts the S3/F35, S4/F36, and S5/F37 breakouts here. First verify S6:T2/F42, whose historical meter-photo control is -20.71 dBm. That control only supports the F42 path; it does not clear the three adjacent branch fibers.

Route Stops and What Each Result Tells You

Coordinates are from the engineering staking KMZ. Enter them in the app/map; do not substitute a nearby network point merely because it has a similar number.

Stop	Staking point / coordinates	Test path	Field hypothesis / next check
1	CP13817 / NP1702 33.365529, -105.674531	F42 control; then S3/F35, S4/F36, S5/F37 before and after breakout	If F42 is good but one of S3-S5 falls across the breakout, suspect that branch splice, port, or outgoing one-count. If the branch is already weak before breakout, move upstream to NP2015.
2	NP2015 - P0002 source 33.367305, -105.673344	P0002 input and outputs S3/F35, S4/F36, S5/F37, S6/F38	Use only when a branch was weak before CP13817 or more than one branch is weak. Compare individual splitter outputs; do not condemn the full feeder from one bad output.
3	NP1696 - S6 distribution 33.366015, -105.673577	S6/F38 in; T1/F41, T2/F42, T3/F43 out; T4 local	Separates a shared S6 feed issue from a single T leg. A bad T3 result points downstream only on T3, not on all of S6.
4	NP1697 - S6:T3 33.365876, -105.674514	S6:T3 / F43	Historical -31.77 dBm. Reverify this exact T3 path. Treat this as separate from S3-S5 until measurements show a common issue.
5	NP2056 / CP11752 33.364960, -105.674550	S5:L1 first PCOT	Historical readings -21.04 and -37.20 dBm, but LOW/HIGH meanings are not defined. Identify actual physical in/out ports and photograph labels before comparison.
6	NP2058 / CP12913 33.364580, -105.673669	S4:L1 first PCOT	Historical readings -39.34 and -22.19 dBm. If current incoming is already weak, the likely interval is CP13817 to this PCOT, not the downstream continuation.
7	NP12001 / CP12917 33.364152, -105.673645	S3:L1 first PCOT	Historical readings -40.29 and -23.17 dBm. Same rule: identify ports first; then compare true incoming to true continuation.

At-the-Closure Test Rules

Use the same wavelength, meter, launch condition, and identified fiber for every comparison. For the app trace, use 1550 nm. Clean and inspect connectors before accepting a surprise number.

1. Establish a trustworthy upstream reading	Measure the known incoming path or source port. A reading around -18 dBm can be normal after prior PON splitting; the important number is the change across the next known interval.
2. Verify physical identity	At a PCOT, do not assume LOW means input or HIGH means output. Photograph enclosure and port labels, record the observed port, and follow the identified fiber.
3. Compare before vs. after	No splitter expected: a small change is normal. A loss of about 3 dB or more across a splice, closure, or short interval is a red flag and justifies inspecting that interval.
4. Account for expected splitter loss	A 1x2 is roughly 3.5-4 dB, 1x4 about 7-8 dB, and 1x8 about 10-11 dB. Do not label expected split loss as a bad splice.
5. Read the branching pattern	If a bad event is before a splitter, all child paths should inherit it. If only one child is weak while neighbors are normal, concentrate on that output, its splice, or its downstream span.
6. Do not let F42 overrule S3-S5	S6:T2/F42 at CP13817 is a control only. It validates its own path, not the neighboring breakout branches.

Quick field log - one identified path per row

Location / path	Physical port ID	In dBm	Out dBm	Loss dB	Expected splitter?	Photo / note

Separate issue: 1687 -> 1688 -> 12801 is 1635CA_04 / P0004 and is intentionally outside this P0002 troubleshooting sequence.

Source basis: Project_NODE54_1635CA.kmz staking data and the Speccom Node 54 P0002 diagnostic configuration. Historical readings are comparison evidence only; LOW/HIGH labels remain undefined.